



# MACHINE LEARNING-DRIVEN BIOMARKER DISCOVERY FOR TUBERCULOSIS

Integrating Multi-Model Learning, Explainable AI, and Biological Interpretation for Blood-Based Biomarker Identification

Abdulrahman S. Hammond | Biomedical Machine Learning Portfolio Project

Tools: Python • Scikit-Learn • XGBoost • SHAP • GEO Datasets • GitHub



## BACKGROUND

Tuberculosis (TB) remains one of the world's leading infectious causes of morbidity and mortality. Accurate blood-based biomarkers could improve diagnosis and support the development of precision medicine approaches.

This study integrates machine learning, explainable AI (SHAP), and biological interpretation to identify robust host-response biomarkers capable of distinguishing active TB from healthy controls.

## OBJECTIVES

- Develop predictive machine-learning models using transcriptomic data
- Identify consensus biomarkers across multiple algorithms
- Apply explainable AI to understand model decisions
- Prioritize biologically meaningful biomarkers for future translational development

## DATASETS

| Dataset  | Purpose             |
|----------|---------------------|
| GSE19491 | Model development   |
| GSE25534 | External validation |

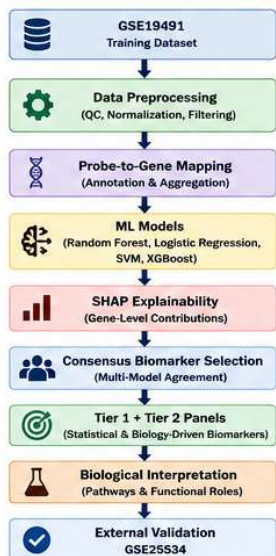


REPOSITORY

[github.com/niisaban/  
biomedical-ml-tb-biomarker-discovery](https://github.com/niisaban/biomedical-ml-tb-biomarker-discovery)

[View Project Repository](#)

## MACHINE LEARNING WORKFLOW



## TABLE B. CONSENSUS BIOMARKER PANELS

| Tier 1 — Multi-Model Consensus Biomarkers |                                    |  |        |
|-------------------------------------------|------------------------------------|--|--------|
| Gene Symbol                               | Model Support Count                |  | Tier   |
| CLEC4G1                                   | 2                                  |  | Tier 1 |
| CPN2                                      | 2                                  |  | Tier 1 |
| DOC2B                                     | 2                                  |  | Tier 1 |
| F7                                        | 2                                  |  | Tier 1 |
| KNSTRN                                    | 2                                  |  | Tier 1 |
| PDEADIP                                   | 2                                  |  | Tier 1 |
| RNA28S5                                   | 2                                  |  | Tier 1 |
| Tier 2 — Biology-Driven Immune Biomarkers |                                    |  |        |
| Gene Symbol                               | Biological Role                    |  |        |
| GBP1P1                                    | Interferon-stimulated response     |  |        |
| NSMAF                                     | TNF / Immune stress signaling      |  |        |
| OSM                                       | Inflammatory cytokine signaling    |  |        |
| PARP14                                    | Immune regulation / STAT signaling |  |        |
| STAT1                                     | Interferon signaling               |  |        |

Tier 1 biomarkers were identified through multi-model machine learning consensus. Tier 2 biomarkers were prioritized based on biological plausibility, immune relevance, and convergence of machine learning and SHAP evidence.

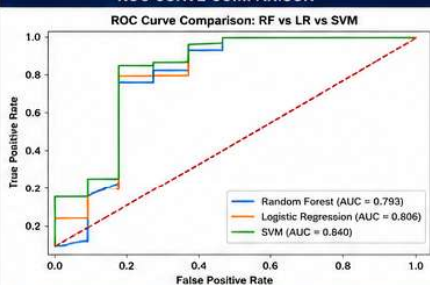
## MODEL PERFORMANCE (Training Dataset: GSE19491)

Table A. Baseline Model Performance

| Model               | Accuracy | Precision | Recall | ROC-AUC |
|---------------------|----------|-----------|--------|---------|
| Logistic Regression | 0.89     | 0.89      | 0.97   | 0.806   |
| Random Forest       | 0.87     | 0.87      | 0.97   | 0.793   |
| SVM                 | NR*      | NR*       | 1.00   | 0.840   |
| XGBoost             | NR*      | NR*       | NR*    | 0.923   |

\*NR = Not Recieved. Metrics were unavailable in archived reports and will be restored following reconstruction of the evaluation notebooks.

## ROC CURVE COMPARISON



All evaluated models exceeded ROC-AUC 0.79, demonstrating consistent discrimination between TB and healthy controls within blood transcriptomic profiles.

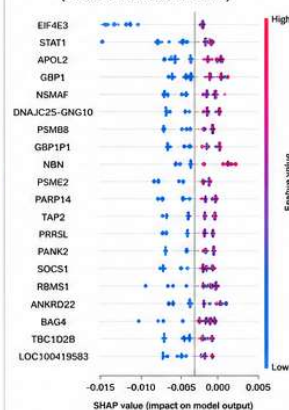
## TABLE C. ROBUSTNESS ASSESSMENT

| Gene    | Tier   | Robustness Assessment |
|---------|--------|-----------------------|
| STAT1   | Tier 2 | High                  |
| PARP14  | Tier 2 | High                  |
| NSMAF   | Tier 2 | High                  |
| GBP1P1  | Tier 2 | High                  |
| OSM     | Tier 2 | Moderate-High         |
| CPN2    | Tier 1 | Moderate              |
| F7      | Tier 1 | Moderate              |
| CLEC4G1 | Tier 1 | Moderate              |
| DOC2B   | Tier 1 | Low-Moderate          |
| PDEADIP | Tier 1 | Low                   |
| KNSTRN  | Tier 1 | Low                   |
| RNA28S5 | Tier 1 | Low                   |

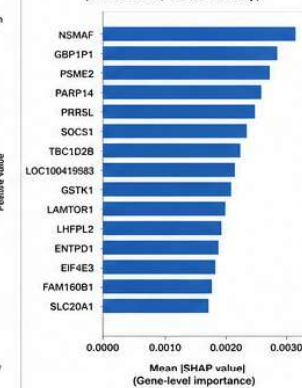
Robustness scores summarize cross-model support, SHAP explainability evidence, and biological consistency. Higher support indicates greater confidence in biomarker reproducibility and downstream translational potential.

## EXPLAINABLE AI (SHAP)

SHAP Summary Plot (Gene-Level Contributions)



Top SHAP Genes (Gene-Level, TB vs Healthy)



SHAP analysis identifies the genes that most strongly influence model predictions, linking computational results to biologically meaningful immune processes.

## TABLE F. BIOLOGICAL INTERPRETATION

| Biomarker | Tier   | Functional Category            | Biological Role                                           |
|-----------|--------|--------------------------------|-----------------------------------------------------------|
| STAT1     | Tier 2 | Interferon Signaling           | Central regulator of IFN-mediated antiviral immunity      |
| GBP1P1    | Tier 2 | Interferon-Stimulated Response | Marker of innate antimicrobial defense activation         |
| PARP14    | Tier 2 | Immune Regulation              | Modulates STAT1-dependent immune signaling pathways       |
| NSMAF     | Tier 2 | TNF Signaling                  | Regulates inflammatory activation and tissue remodeling   |
| OSM       | Tier 2 | Cytokine Signaling             | Associated with systemic inflammatory responses           |
| CPN2      | Tier 2 | Complement/Memecostasis        | Associated with systemic inflammatory responses           |
| F7        | Tier 1 | Coagulation Pathways           | Links inflammation and coagulation processes              |
| CLEC4G1   | Tier 1 | Lectin-Associated Immunity     | Immune recognition and signaling                          |
| DOC2B     | Tier 1 | Cellular Signaling             | Structural biomarker with limited cellular processes      |
| PDEADIP   | Tier 1 | Signal Transduction            | Structural and signaling-associated cellular processes    |
| KNSTRN    | Tier 1 | Cell Cycle Regulation          | Statistical biomarker with limited biological evidence    |
| RNA28S5   | Tier 1 | Ribosomal RNA Component        | Statistical biomarker with limited functional TB evidence |

Functional annotations were derived from machine-learning feature selection, SHAP explainability analyses, pathway interpretation, and published TB immunology literature.

## CONCLUSIONS



Multi-model machine learning identified reproducible TB-associated biomarkers.



Explainable AI linked predictive features to known immune pathways.



Tier 2 biomarkers demonstrated the highest translational potential.



Consensus biomarker selection improved confidence in candidate-gene prioritization.



XGBoost achieved the strongest predictive performance (AUC = 0.923).

Top-ranked biomarkers converged on interferon, cytokine, and TNF-associated immune pathways known to play central roles in host responses to *Mycobacterium tuberculosis*.

## FUTURE WORK

- Reconstruct archived notebooks to recover missing evaluation metrics.
- Complete external validation on GSE25534.
- Evaluate Tier 1 vs Tier 2 panel performance.
- Expand pathway and network-level analyses.
- Develop a deployable biomarker-screening dashboard.

## KEYWORDS

Tuberculosis  
Biomarkers  
Machine Learning  
Explainable AI  
SHAP  
Transcriptomics  
Precision Medicine